

Analysis PhD Exam, January 2019

Do three from part A and three from part B. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

Part A

1. Let X, Y be topological spaces. Prove that every continuous function $f : X \rightarrow Y$ is Borel measurable.
2. Short answer.
 - (a) Give an example, if possible, of a closed set $E \subset \mathbb{R}$ of positive Lebesgue measure that contains no interval.
 - (b) For $n \geq 2$, let $f_n = e^{inx}x^{-n}$. Find

$$\lim_{n \rightarrow \infty} \int_1^\infty f_n dx.$$

- (c) Give an example, if possible, of strict inequality in Fatou's Lemma.
3. Let $X = Y = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$ denote the Borel σ -algebra, and $\mathcal{N} = 2^{\mathbb{R}}$. Let μ Lebesgue measure on $\mathcal{M}$, and let ν counting measure on $\mathcal{N}$. Prove that $\Delta = \{(x, x) : x \in [0, 1]\} \subset [0, 1] \times [0, 1]$ is an element of the product σ -algebra $\mathcal{M} \otimes \mathcal{N}$ and compute

$$\int_X \left(\int_Y \mathbf{1}_\Delta(x, y) d\nu(y) \right) d\mu(x), \quad \int_Y \left(\int_X \mathbf{1}_\Delta(x, y) d\mu(x) \right) d\nu(y). \quad (1)$$

Explain the relation to Tonelli's Theorem on product integration.

4. State the Hahn Decomposition Theorem. Prove, if ρ is a signed measure on the measurable space $(X, \mathcal{M})$, then there exist unique positive measures $\rho_\pm$ on $\mathcal{M}$ such that $\rho_+ \perp \rho_-$ and $\rho = \rho_+ - \rho_-$.
Assuming μ is a positive measure on $\mathcal{M}$ and $f \in L^1(\mu)$ is real-valued, identify $\rho_\pm$ for the signed measure

$$\rho(E) = \int_E f d\mu.$$

Part B

5. State the Baire Category Theorem. Suppose X is a Banach space. Prove that, as a vector space, X does not have a countable basis. (Here countable means equinumerous with $\mathbb{N}$.)
6. Let (φ_n) be a sequence from $L^1(\mathbb{R})$ with the following properties:
Show, if $f \in L^1(\mathbb{R})$, then $\varphi_n \star f$ converges to f in $L^1(\mathbb{R})$.

- (i) $\varphi_n \geq 0$ for all n ,
- (ii) $\int \varphi_n dx = 1$ for all n ;
- (iii) for each $\delta > 0$, $\lim_{n \rightarrow \infty} \int_{|x| > \delta} \varphi_n dx = 0$.

7. Let X be a Banach space, $Y \subset X$ a closed subspace. Say Y is complemented in X if there exists a closed subspace $Z \subset X$ such that $Y \cap Z = \{0\}$ and $Y + Z = X$.

Prove that if Y is complemented in X , then there exists a bounded linear operator $T : X \rightarrow Y$ such that $T(y) = y$ for all $y \in Y$.

8. Let $S : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ denote the shift operator defined, for $f = (f(n))_{n=0}^\infty \in \ell^2(\mathbb{N})$, by

$$Sf(n) = \begin{cases} f(n-1), & n \geq 1; \\ 0, & n = 0. \end{cases}$$

Show S is an isometry (and in particular is bounded) and find S^* and SS^* . Is S unitary?